ADAR'S STUDY SERIES

Linear Algebra

Part 4

The Matrix Equation $A\vec{x} = \vec{b}$
& Solution Sets in Parametric Vector Form

Contents

PART 4A

Matrix Equation $A\vec{x} = \vec{b}$

Overview & Goals

- **Goal 1:** Learn how to compute $A\vec{x}$ in 2 ways.
- **Goal 2:** Understand how to write a linear system in **vector form** and **matrix form**.

Defining Matrix–Vector Multiplication $A\vec{x}$

Definition: Let $A = [\ \vec{a}_1\ \ \vec{a}_2\ \ \ldots\ \ \vec{a}_n]$ be an $m \times n$ matrix (m rows, n columns), where each column vector $\vec{a}_i \in \mathbb{R}^m$.

Let $\vec{x}$ =

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

$\in \mathbb{R}^n$.

Then the product $A\vec{x}$ is defined as a **linear combination of the columns of A**:

$$A\vec{x} = \begin{bmatrix} \vec{a}_1 & \vec{a}_2 & \ldots & \vec{a}_n \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

$$:= x_1\vec{a}_1 + x_2\vec{a}_2 + \ldots + x_n\vec{a}_n$$

PART 4A

Example 1: Computing $A\vec{x}$

Compute the following, if defined:

Part (a)

Compute

$$\begin{bmatrix} 0 & -3 & 5 \\ 2 & 1 & 4 \end{bmatrix} \begin{bmatrix} 5 \\ -1 \\ 2 \end{bmatrix}$$

- **Method 1: Linear Combination of Columns**

$$= 5 \begin{bmatrix} 0 \\ 2 \end{bmatrix} - 1 \begin{bmatrix} -3 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 5 \\ 4 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 10 \end{bmatrix} + \begin{bmatrix} 3 \\ -1 \end{bmatrix} + \begin{bmatrix} 10 \\ 8 \end{bmatrix} = \begin{bmatrix} \mathbf{13} \\ \mathbf{17} \end{bmatrix}$$

- **Method 2: Row-Vector Rule (Dot Product Rule)**

$$\begin{bmatrix} 5(0) + (-1)(-3) + 2(5) \\ 5(2) + (-1)(1) + 2(4) \end{bmatrix} = \begin{bmatrix} \mathbf{13} \\ \mathbf{17} \end{bmatrix}$$

Part (b)

Compute

$$\begin{bmatrix} 0 & -3 & 5 \\ 2 & 1 & 4 \end{bmatrix} \begin{bmatrix} 5 \\ -1 \end{bmatrix}$$

- **Result:** Undefined.
- **Reason:** The number of columns of $A(3)$ does not match the number of entries in vector $\vec{x}(2)$.

Part (c)

Compute

$$\begin{bmatrix} 1 & -2 & 1 \\ 0 & 0 & 5 \\ 0 & 3 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ 0 \\ 7 \end{bmatrix}$$

- **Using the Row-Vector Rule:**

$$= \begin{bmatrix} 3(1) + 0(-2) + 7(1) \\ 3(0) + 0(0) + 7(5) \\ 3(0) + 0(3) + 7(-1) \end{bmatrix} = \begin{bmatrix} \mathbf{10} \\ \mathbf{35} \\ \mathbf{-7} \end{bmatrix}$$

PART 4A

Example 2: Systems of Equations in Vector and Matrix Form

Part (a): Converting Systems

Given the system of linear equations:

$$6x_1 - x_2 = 9$$
$$4x_1 + 3x_2 = -5$$
$$x_1 + 7x_2 = -20$$

- **Vector Form:**

$$x_1 \begin{bmatrix} 6 \\ 4 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} -1 \\ 3 \\ 7 \end{bmatrix} = \begin{bmatrix} 9 \\ -5 \\ -20 \end{bmatrix}$$

- **Matrix Form ($A\vec{x} = \vec{b}$):**

$$\begin{bmatrix} 6 & -1 \\ 4 & 3 \\ 1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 9 \\ -5 \\ -20 \end{bmatrix}$$

Part (b): Solving the System

Solve the matrix equation $A\vec{x} = \vec{b}$ using row reduction:

1. **Augmented Matrix:**

$$\left[\begin{array}{cc|c} 6 & -1 & 9 \\ 4 & 3 & -5 \\ 1 & 7 & -20 \end{array}\right]$$

2. **Swap Rows ($R_1 \leftrightarrow R_3$):**

$$\Rightarrow \left[\begin{array}{cc|c} 1 & 7 & -20 \\ 4 & 3 & -5 \\ 6 & -1 & 9 \end{array}\right]$$

3. **Eliminate lower column 1 entries:**

    - $-4R_1 + R_2 \to R_2$
    - $-6R_1 + R_3 \to R_3$

$$\Rightarrow \left[\begin{array}{cc|c} 1 & 7 & -20 \\ 0 & -25 & 75 \\ 0 & -43 & 129 \end{array}\right]$$

4. **Rescale Rows:**

    - $R_2\,(-{}^{1}\!/_{25}) \to R_2$
    - $R_3\,(-{}^{1}\!/_{43}) \to R_3$

$$\Rightarrow \left[\begin{array}{cc|c} 1 & 7 & -20 \\ 0 & 1 & -3 \\ 0 & 1 & -3 \end{array}\right]$$

5. **Eliminate Row 3:**

    - $R_3 - R_2 \to R_3$

$$\Rightarrow \left[\begin{array}{cc|c} 1 & 7 & -20 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{array}\right]$$

6. **Reduced Row Echelon Form (RREF):**

   - $-7R_2 + R_1 \rightarrow R_1$

$$\Longrightarrow \left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{array}\right]$$

- **Final Solution:**

$$x_1 = 1$$
$$x_2 = -3$$

$$\Longrightarrow \quad \vec{x} \; = \begin{bmatrix} 1 \\ -3 \end{bmatrix}$$

Solution Sets of Linear Systems

Overview & Goals

- **Goal 1:** Learn how to write the general solution to a linear system in **parametric vector form**.
- **Goal 2:** Understand geometrically the difference in the solution sets of $A\vec{x} = \vec{0}$ and $A\vec{x} = \vec{b}$ when $\vec{b} \neq \vec{0}$.

PART 4B

Homogeneous Linear Systems

Definition: A linear system is called **homogeneous** if it can be written in the form:

$$A\vec{x} = \vec{0}$$

where A is an $m \times n$ matrix and $\vec{0}$ is the zero vector in $\mathbb{R}^m$.

In terms of columns of A:

$$A\vec{x} = \vec{0} \Leftrightarrow x_1\vec{a}_1 + x_2\vec{a}_2 + \ldots + x_n\vec{a}_n = \vec{0}$$

Note: The vector $\vec{x} = \vec{0}$ (i.e., $x_1 = x_2 = \ldots = x_n = 0$) is **always** a solution to a homogeneous system. This is called the **trivial solution**. Nonzero solutions, if they exist, are called **nontrivial solutions**.

Example 1: Homogeneous System

Find the solution set of the following system in **parametric vector form**:

$$x_3 + 4x_4 - x_5 = 0$$
$$2x_1 - 2x_2 + 4x_3 + 6x_4 + 3x_5 = 0$$
$$x_1 - x_2 + x_3 - x_4 + 2x_5 = 0$$

Solution Steps:

1. **Augmented Matrix & Row Reduction to RREF:**

$$\left[\begin{array}{ccccc|c} 0 & 0 & 1 & 4 & -1 & 0 \\ 2 & -2 & 4 & 6 & 3 & 0 \\ 1 & -1 & 1 & -1 & 2 & 0 \end{array}\right] \sim \left[\begin{array}{ccccc|c} 1 & -1 & 0 & -5 & 0 & 0 \\ 0 & 0 & 1 & 4 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{array}\right]$$

2. **Identify Variables:**
   - **Pivot Variables:** x_1, x_3, x_5
   - **Free Variables:** x_2, x_4

3. **Express Pivot Variables in Terms of Free Variables:**

$$x_1 - x_2 - 5x_4 = 0 \implies x_1 = x_2 + 5x_4$$
$$x_3 + 4x_4 = 0 \implies x_3 = -4x_4$$
$$x_5 = 0 \implies x_5 = 0$$

Writing every variable explicitly:

$$x_1 = x_2 + 5x_4$$
$$x_2 = free$$
$$x_3 = -4x_4$$
$$x_4 = free$$
$$x_5 = 0$$

4. **General Solution in Parametric Vector Form:**

$$\vec{x} =$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} x_2 + 5x_4 \\ x_2 \\ -4x_4 \\ x_4 \\ 0 \end{bmatrix}$$

$$= x_2 \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 5 \\ 0 \\ -4 \\ 1 \\ 0 \end{bmatrix}$$

(This is the final parametric vector form)

PART 4B

Nonhomogeneous Linear Systems

Definition: A linear system is called **nonhomogeneous** if it can be written in the form:

$$A\vec{x} = \vec{b} \quad \text{where } \vec{b} \neq \vec{0}$$

Example 2: Nonhomogeneous System

Find the solution set of the following system in **parametric vector form**:

$$x_3 + 4x_4 - x_5 = 1$$
$$2x_1 - 2x_2 + 4x_3 + 6x_4 + 3x_5 = 7$$
$$x_1 - x_2 + x_3 - x_4 + 2x_5 = 1$$

Solution Steps:

1. **Augmented Matrix & Row Reduction to RREF:**

$$\left[\begin{array}{ccccc|c} 0 & 0 & 1 & 4 & -1 & 1 \\ 2 & -2 & 4 & 6 & 3 & 7 \\ 1 & -1 & 1 & -1 & 2 & 1 \end{array}\right] \sim \left[\begin{array}{ccccc|c} 1 & -1 & 0 & -5 & 0 & -9 \\ 0 & 0 & 1 & 4 & 0 & 4 \\ 0 & 0 & 0 & 0 & 1 & 3 \end{array}\right]$$

2. **Express Pivot Variables in Terms of Free Variables:**

$$x_1 - x_2 - 5x_4 = -9 \implies x_1 = -9 + x_2 + 5x_4$$
$$x_3 + 4x_4 = 4 \implies x_3 = 4 - 4x_4$$
$$x_5 = 3 \implies x_5 = 3$$

Writing every variable explicitly:

$$x_1 = -9 + x_2 + 5x_4$$
$$x_2 = free$$
$$x_3 = 4 - 4x_4$$
$$x_4 = free$$
$$x_5 = 3$$

3. General Solution in Parametric Vector Form:

$\vec{x} =$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -9 + x_2 + 5x_4 \\ x_2 \\ 4 - 4x_4 \\ x_4 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} -9 \\ 0 \\ 4 \\ 0 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 5 \\ 0 \\ -4 \\ 1 \\ 0 \end{bmatrix}$$

(This is the final parametric vector form)

PART 4B

Solution Structure Theorem

Notice how the solution vector $\vec{x}$ to $A\vec{x} = \vec{b}$ splits into two distinct components:

$$\vec{x} = \begin{bmatrix} -9 \\ 0 \\ 4 \\ 0 \\ 3 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 5 \\ 0 \\ -4 \\ 1 \\ 0 \end{bmatrix}$$

$$-- \vec{v}_p -- \qquad -------\vec{v}_h-------$$

- $\vec{v}_p$: A **particular solution** to $A\vec{x} = \vec{b}$ (satisfies $A\vec{v}_p = \vec{b}$).
- $\vec{v}_h$: The **homogeneous solution** (the general solution to $A\vec{x} = \vec{0}$).

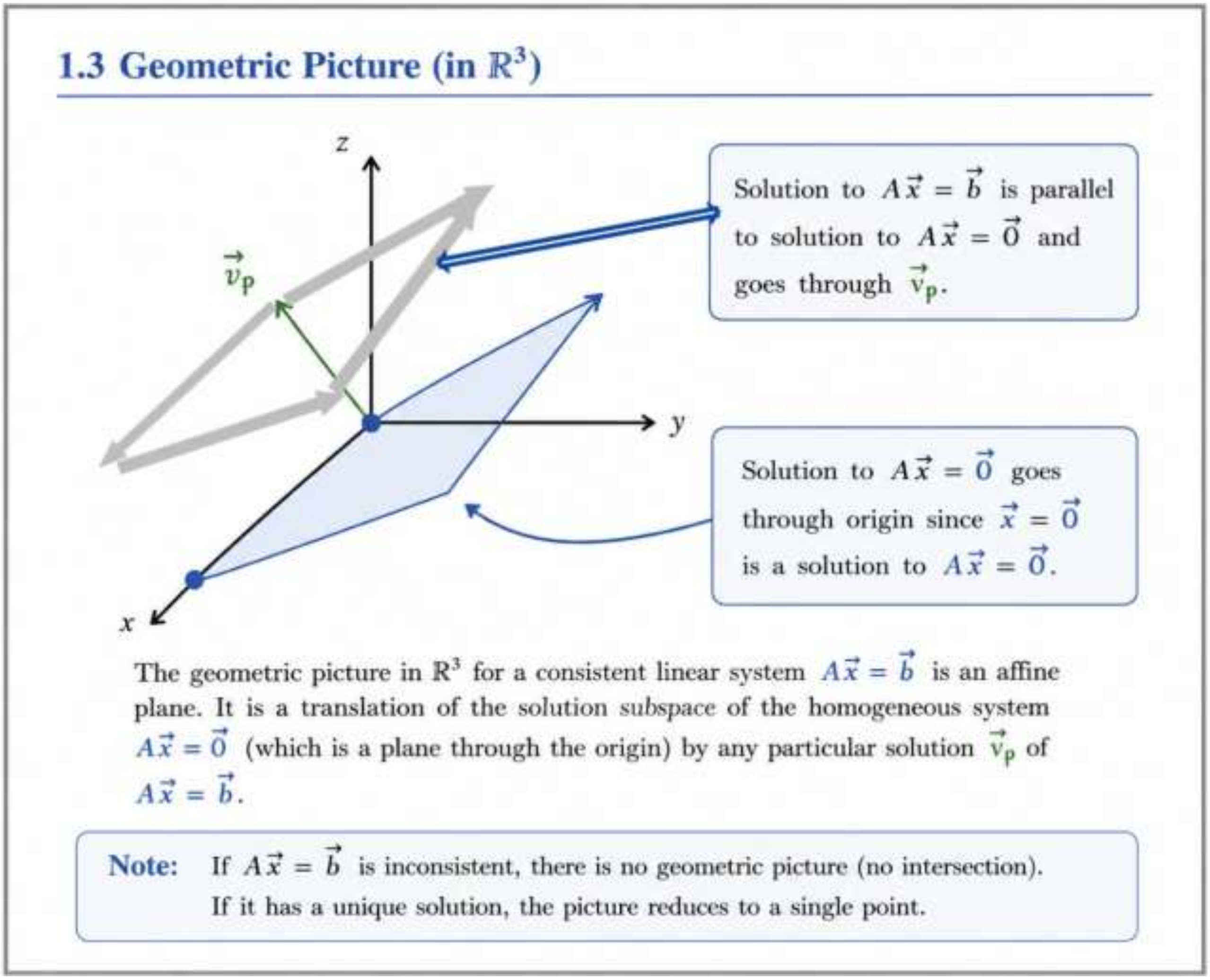
Theorem: Suppose the equation $A\vec{x} = \vec{b}$ is consistent for a given vector $\vec{b}$, and let $\vec{v}_p$ be a particular solution. Then the solution set of $A\vec{x} = \vec{b}$ (where $\vec{b} \neq \vec{0}$) is the set of all vectors of the form:

$$\vec{x} = \vec{v}_p + \vec{v}_h$$

where $\vec{v}_h$ is any solution of the corresponding homogeneous system $A\vec{x} = \vec{0}$.

PART 4B

Geometric Picture (in $\mathbb{R}^3$)



The solution set of $A\vec{x} = \vec{b}$ is **parallel** to the solution set of $A\vec{x} = \vec{0}$, and passes through $\vec{v}_p$. The solution set of $A\vec{x} = \vec{0}$ passes through the origin, since $\vec{x} = \vec{0}$ is always a solution to $A\vec{x} = \vec{0}$.

END OF PART 3 · NEXT: PART 4 — LINEAR INDEPENDENCE